PRAKTIKUM BASIS DATA

MODUL 03

TUGAS PENDAHULUAN

Oleh:

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

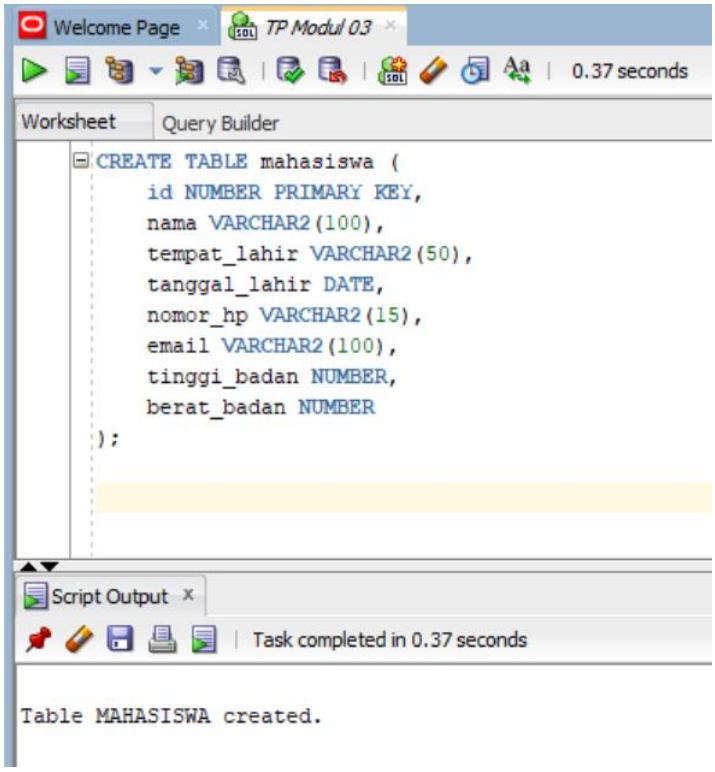
UNIVERSITAS TELKOM

2026

JURNAL/TP

1. **Buat Table Mahasiswa yang beratribut (id, nama, tempat lahir, tanggal lahir, nomor hp, email, tinggi badan, berat badan.)**

   Jawaban:



2. **Lakukan insert data pada table nomor 1 sebanyak 15 rows.**

Jawaban:

Welcome Page | TP Modul 03 | 0.213 seconds | TP Modul 03

Worksheet | Query Builder

```
INSERT INTO mahasiswa VALUES (1,'Aldi Prayoga','Jakarta',TO_DATE('2003-01-12','YYYY-MM-
INSERT INTO mahasiswa VALUES (2,'Budiman Santoso','Bandung',TO_DATE('2002-05-20','YYYY-
INSERT INTO mahasiswa VALUES (3,'Ciara Putri','Surabaya',TO_DATE('2003-07-15','YYYY-MM-
INSERT INTO mahasiswa VALUES (4,'Dodi Nugroho','Semarang',TO_DATE('2001-03-18','YYYY-MM
INSERT INTO mahasiswa VALUES (5,'Ekantha Putri','Yogyakarta',TO_DATE('2002-09-25','YYYY
INSERT INTO mahasiswa VALUES (6,'Firman Abdul','Malang',TO_DATE('2003-11-11','YYYY-MM-D
INSERT INTO mahasiswa VALUES (7,'Gira Firmansyah','Bogor',TO_DATE('2002-02-02','YYYY-MM
INSERT INTO mahasiswa VALUES (8,'Haryanto Kurniawan','Solo',TO_DATE('2001-12-30','YYYY-
INSERT INTO mahasiswa VALUES (9,'Indah Permata','Medan',TO_DATE('2003-04-09','YYYY-MM-D
INSERT INTO mahasiswa VALUES (10,'Jono Susilo','Palembang',TO_DATE('2002-08-14','YYYY-M
INSERT INTO mahasiswa VALUES (11,'Kiana Nanda','Denpasar',TO_DATE('2003-06-06','YYYY-MM
INSERT INTO mahasiswa VALUES (12,'Lukman Hakim','Makassar',TO_DATE('2001-10-01','YYYY-M
INSERT INTO mahasiswa VALUES (13,'Meylani Sari','Padang',TO_DATE('2002-03-21','YYYY-MM-
INSERT INTO mahasiswa VALUES (14,'Nanda Hakim','Pontianak',TO_DATE('2003-09-17','YYYY-M
INSERT INTO mahasiswa VALUES (15,'Oki Purwanto','Balikpapan',TO_DATE('2002-12-05','YYYY
```

Script Output | Task completed in 0.213 seconds

```
1 row inserted.
```

Welcome Page | TP Modul 03

Worksheet | Query Builder

```
INSERT INTO mahasiswa VALUES (1,'Aldi Prayoga','Jakarta',TO_DATE('2003-01-12','YYYY-MM-DD'),'081234567001','aldi@gmail.com',170,65);
INSERT INTO mahasiswa VALUES (2,'Budiman Santoso','Bandung',TO_DATE('2002-05-20','YYYY-MM-DD'),'081234567002','budiman@gmail.com',168,60);
INSERT INTO mahasiswa VALUES (3,'Ciara Putri','Surabaya',TO_DATE('2003-07-15','YYYY-MM-DD'),'081234567003','ciara@gmail.com',160,50);
INSERT INTO mahasiswa VALUES (4,'Dodi Nugroho','Semarang',TO_DATE('2001-03-18','YYYY-MM-DD'),'081234567004','dodi@gmail.com',172,68);
INSERT INTO mahasiswa VALUES (5,'Ekantha Putri','Yogyakarta',TO_DATE('2002-09-25','YYYY-MM-DD'),'081234567005','ekanatha@gmail.com',158,48);
INSERT INTO mahasiswa VALUES (6,'Firman Abdul','Malang',TO_DATE('2003-11-11','YYYY-MM-DD'),'081234567006','firman@gmail.com',175,70);
INSERT INTO mahasiswa VALUES (7,'Gira Firmansyah','Bogor',TO_DATE('2002-02-02','YYYY-MM-DD'),'081234567007','gira@gmail.com',162,52);
INSERT INTO mahasiswa VALUES (8,'Haryanto Kurniawan','Solo',TO_DATE('2001-12-30','YYYY-MM-DD'),'081234567008','haryanto@gmail.com',169,66);
INSERT INTO mahasiswa VALUES (9,'Indah Permata','Medan',TO_DATE('2003-04-09','YYYY-MM-DD'),'081234567009','indah@gmail.com',159,49);
INSERT INTO mahasiswa VALUES (10,'Jono Susilo','Palembang',TO_DATE('2002-08-14','YYYY-MM-DD'),'081234567010','jono@gmail.com',173,72);
INSERT INTO mahasiswa VALUES (11,'Kiana Nanda','Denpasar',TO_DATE('2003-06-06','YYYY-MM-DD'),'081234567011','kiana@gmail.com',161,51);
INSERT INTO mahasiswa VALUES (12,'Lukman Hakim','Makassar',TO_DATE('2001-10-01','YYYY-MM-DD'),'081234567012','lukman@gmail.com',174,69);
INSERT INTO mahasiswa VALUES (13,'Meylani Sari','Padang',TO_DATE('2002-03-21','YYYY-MM-DD'),'081234567013','meylani@gmail.com',157,47);
INSERT INTO mahasiswa VALUES (14,'Nanda Hakim','Pontianak',TO_DATE('2003-09-17','YYYY-MM-DD'),'081234567014','nanda@gmail.com',171,67);
INSERT INTO mahasiswa VALUES (15,'Oki Purwanto','Balikpapan',TO_DATE('2002-12-05','YYYY-MM-DD'),'081234567015','oki@gmail.com',176,73);

SELECT * FROM mahasiswa;
```

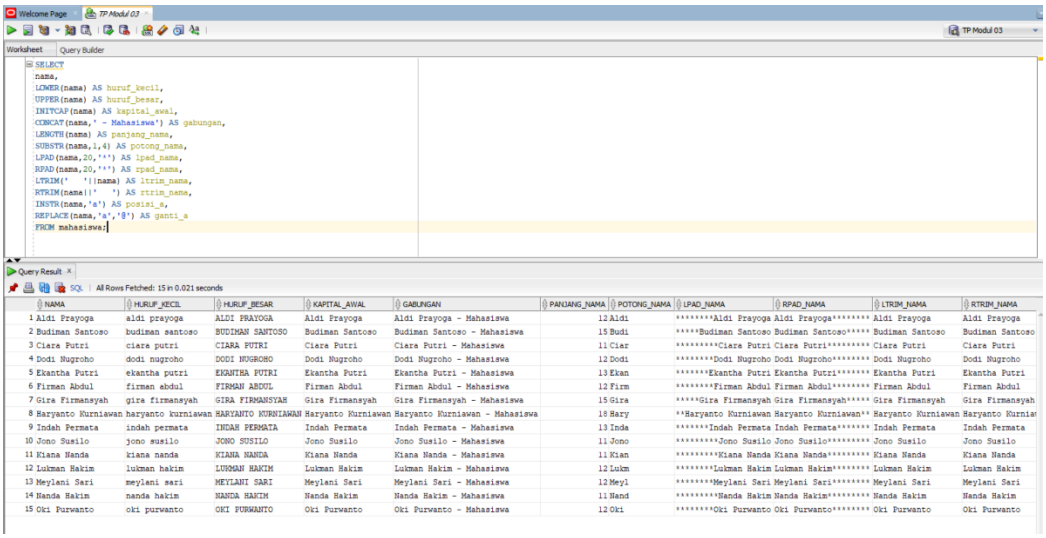
Query Result | SQL | All Rows Fetched: 15 in 0.068 seconds

	ID	NAMA	TEMPAT_LAHIR	TANGGAL_LAHIR	NOMOR_HP	EMAIL	TINGGI_BADAN	BERAT_BADAN
1	1	Aldi Prayoga	Jakarta	12-01-2003	081234567001	aldi@gmail.com	170	65
2	2	Budiman Santoso	Bandung	20-05-2002	081234567002	budiman@gmail.com	168	60
3	3	Ciara Putri	Surabaya	15-07-2003	081234567003	ciara@gmail.com	160	50
4	4	Dodi Nugroho	Semarang	18-03-2001	081234567004	dodi@gmail.com	172	68
5	5	Ekantha Putri	Yogyakarta	25-09-2002	081234567005	ekanatha@gmail.com	158	48
6	6	Firman Abdul	Malang	11-11-2003	081234567006	firman@gmail.com	175	70
7	7	Gira Firmansyah	Bogor	02-02-2002	081234567007	gira@gmail.com	162	52
8	8	Haryanto Kurniawan	Solo	30-12-2001	081234567008	haryanto@gmail.com	169	66
9	9	Indah Permata	Medan	09-04-2003	081234567009	indah@gmail.com	159	49
10	10	Jono Susilo	Palembang	14-08-2002	081234567010	jono@gmail.com	173	72
11	11	Kiana Nanda	Denpasar	06-06-2003	081234567011	kiana@gmail.com	161	51
12	12	Lukman Hakim	Makassar	01-10-2001	081234567012	lukman@gmail.com	174	69
13	13	Meylani Sari	Padang	21-03-2002	081234567013	meylani@gmail.com	157	47
14	14	Nanda Hakim	Pontianak	17-09-2003	081234567014	nanda@gmail.com	171	67
15	15	Oki Purwanto	Balikpapan	05-12-2002	081234567015	oki@gmail.com	176	73

3. Lakukan Fungsi

a. Fungsi Karakter dan String

Jawaban :



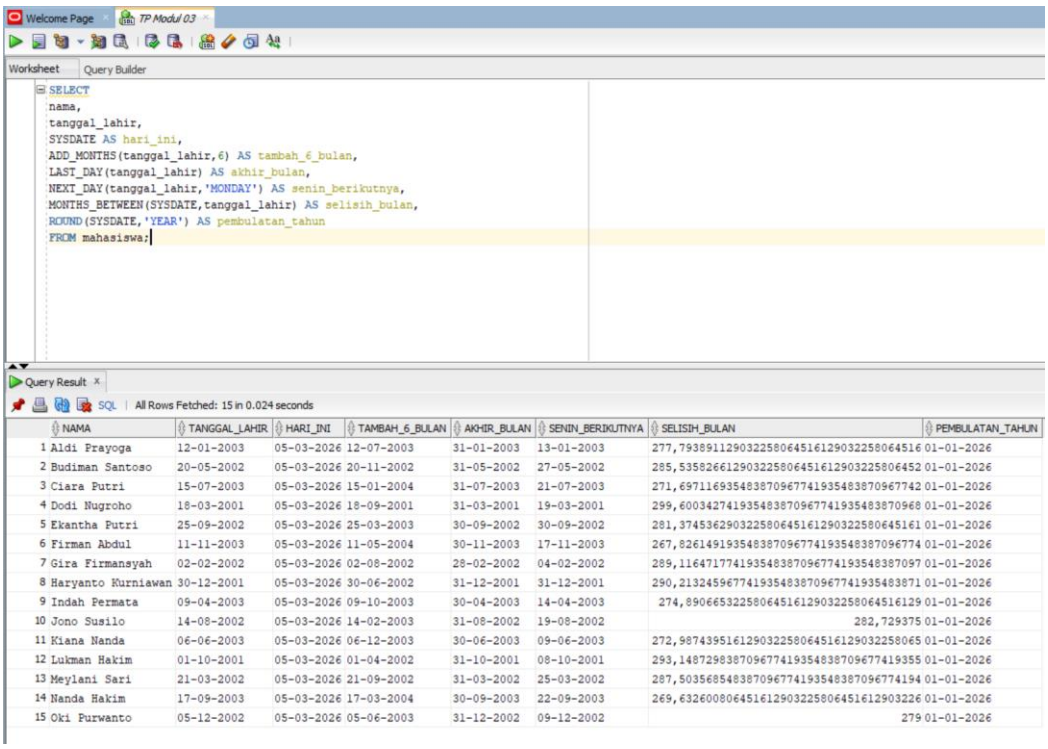
```
SELECT
nama,
LOWER(nama) AS huruf_kecil,
UPPER(nama) AS huruf_besar,
INITCAP(nama) AS kapital_awal,
CONCAT(nama,' - Mahasiswa') AS gabungan,
LENGTH(nama) AS panjang_nama,
SUBSTR(nama,1,4) AS potong_nama,
LPAD(nama,20,'*') AS lpad_nama,
RPAD(nama,20,'*') AS rpad_nama,
LTRIM('  '||nama) AS ltrim_nama,
RTRIM(nama||'  ') AS rtrim_nama,
INSTR(nama,'a') AS posisi_a,
REPLACE(nama,'a','8') AS ganti_a
FROM mahasiswa;
```

Query Result — All Rows Fetched: 15 in 0.021 seconds

	NAMA	HURUF_KECIL	HURUF_BESAR	KAPITAL_AWAL	GABUNGAN	PANJANG_NAMA	POTONG_NAMA	LPAD_NAMA	RPAD_NAMA	LTRIM_NAMA	RTRIM_NAMA
1	Aldi Prayoga	aldi prayoga	ALDI PRAYOGA	Aldi Prayoga	Aldi Prayoga - Mahasiswa	12	Aldi	********Aldi Prayoga	Aldi Prayoga********	Aldi Prayoga	Aldi Prayoga
2	Budiman Santoso	budiman santoso	BUDIMAN SANTOSO	Budiman Santoso	Budiman Santoso - Mahasiswa	15	Budi	*****Budiman Santoso	Budiman Santoso*****	Budiman Santoso	Budiman Santoso
3	Ciara Putri	ciara putri	CIARA PUTRI	Ciara Putri	Ciara Putri - Mahasiswa	11	Ciar	*********Ciara Putri	Ciara Putri*********	Ciara Putri	Ciara Putri
4	Dodi Nugroho	dodi nugroho	DODI NUGROHO	Dodi Nugroho	Dodi Nugroho - Mahasiswa	12	Dodi	********Dodi Nugroho	Dodi Nugroho********	Dodi Nugroho	Dodi Nugroho
5	Ekantha Putri	ekantha putri	EKANTHA PUTRI	Ekantha Putri	Ekantha Putri - Mahasiswa	13	Ekan	*******Ekantha Putri	Ekantha Putri*******	Ekantha Putri	Ekantha Putri
6	Firman Abdul	firman abdul	FIRMAN ABDUL	Firman Abdul	Firman Abdul - Mahasiswa	12	Firm	********Firman Abdul	Firman Abdul********	Firman Abdul	Firman Abdul
7	Gira Firmansyah	gira firmansyah	GIRA FIRMANSYAH	Gira Firmansyah	Gira Firmansyah - Mahasiswa	15	Gira	*****Gira Firmansyah	Gira Firmansyah*****	Gira Firmansyah	Gira Firmansyah
8	Haryanto Kurniawan	haryanto kurniawan	HARYANTO KURNIAWAN	Haryanto Kurniawan	Haryanto Kurniawan - Mahasiswa	18	Hary	**Haryanto Kurniawan	Haryanto Kurniawan**	Haryanto Kurniawan	Haryanto Kurnia
9	Indah Permata	indah permata	INDAH PERMATA	Indah Permata	Indah Permata - Mahasiswa	13	Inda	*******Indah Permata	Indah Permata*******	Indah Permata	Indah Permata
10	Jono Susilo	jono susilo	JONO SUSILO	Jono Susilo	Jono Susilo - Mahasiswa	11	Jono	*********Jono Susilo	Jono Susilo*********	Jono Susilo	Jono Susilo
11	Kiana Nanda	kiana nanda	KIANA NANDA	Kiana Nanda	Kiana Nanda - Mahasiswa	11	Kian	*********Kiana Nanda	Kiana Nanda*********	Kiana Nanda	Kiana Nanda
12	Lukman Hakim	lukman hakim	LUKMAN HAKIM	Lukman Hakim	Lukman Hakim - Mahasiswa	12	Lukm	********Lukman Hakim	Lukman Hakim********	Lukman Hakim	Lukman Hakim
13	Meylani Sari	meylani sari	MEYLANI SARI	Meylani Sari	Meylani Sari - Mahasiswa	12	Meyl	********Meylani Sari	Meylani Sari********	Meylani Sari	Meylani Sari
14	Nanda Hakim	nanda hakim	NANDA HAKIM	Nanda Hakim	Nanda Hakim - Mahasiswa	11	Nand	********Nanda Hakim	Nanda Hakim********	Nanda Hakim	Nanda Hakim
15	Oki Purwanto	oki purwanto	OKI PURWANTO	Oki Purwanto	Oki Purwanto - Mahasiswa	12	Oki	********Oki Purwanto	Oki Purwanto********	Oki Purwanto	Oki Purwanto

b. Fungsi Tanggal dan Waktu

Jawaban :



```
SELECT
nama,
tanggal_lahir,
SYSDATE AS hari_ini,
ADD_MONTHS(tanggal_lahir,6) AS tambah_6_bulan,
LAST_DAY(tanggal_lahir) AS akhir_bulan,
NEXT_DAY(tanggal_lahir,'MONDAY') AS senin_berikutnya,
MONTHS_BETWEEN(SYSDATE,tanggal_lahir) AS selisih_bulan,
ROUND(SYSDATE,'YEAR') AS pembulatan_tahun
FROM mahasiswa;
```

Query Result — All Rows Fetched: 15 in 0.024 seconds

	NAMA	TANGGAL_LAHIR	HARI_INI	TAMBAH_6_BULAN	AKHIR_BULAN	SENIN_BERIKUTNYA	SELISIH_BULAN	PEMBULATAN_TAHUN
1	Aldi Prayoga	12-01-2003	05-03-2026	12-07-2003	31-01-2003	13-01-2003	277,79389112903225806451612903225806451 6	01-01-2026
2	Budiman Santoso	20-05-2002	05-03-2026	20-11-2002	31-05-2002	27-05-2002	285,53582661290322580645161290322580645 2	01-01-2026
3	Ciara Putri	15-07-2003	05-03-2026	15-01-2004	31-07-2003	21-07-2003	271,69711693548387096774193548387096774 2	01-01-2026
4	Dodi Nugroho	18-03-2001	05-03-2026	18-09-2001	31-03-2001	19-03-2001	299,60034274193548387096774193548387096 8	01-01-2026
5	Ekantha Putri	25-09-2002	05-03-2026	25-03-2003	30-09-2002	30-09-2002	281,37453629032258064516129032258064516 1	01-01-2026
6	Firman Abdul	11-11-2003	05-03-2026	11-05-2004	30-11-2003	17-11-2003	267,82614919354838709677419354838709677 4	01-01-2026
7	Gira Firmansyah	02-02-2002	05-03-2026	02-08-2002	28-02-2002	04-02-2002	289,11647177419354838709677419354838709 7	01-01-2026
8	Haryanto Kurniawan	30-12-2001	05-03-2026	30-06-2002	31-12-2001	31-12-2001	290,21324596774193548387096774193548387 1	01-01-2026
9	Indah Permata	09-04-2003	05-03-2026	09-10-2003	30-04-2003	14-04-2003	274,89066532258064516129032258064516129	01-01-2026
10	Jono Susilo	14-08-2002	05-03-2026	14-02-2003	31-08-2002	19-08-2002	282,729375	01-01-2026
11	Kiana Nanda	06-06-2003	05-03-2026	06-12-2003	30-06-2003	09-06-2003	272,98743951612903225806451612903225806 5	01-01-2026
12	Lukman Hakim	01-10-2001	05-03-2026	01-04-2002	31-10-2001	08-10-2001	293,14872983870967741935483870967741935 5	01-01-2026
13	Meylani Sari	21-03-2002	05-03-2026	21-09-2002	31-03-2002	25-03-2002	287,50356854838709677419354838709677419 4	01-01-2026
14	Nanda Hakim	17-09-2003	05-03-2026	17-03-2004	30-09-2003	22-09-2003	269,63260080645161290322580645161290322 6	01-01-2026
15	Oki Purwanto	05-12-2002	05-03-2026	05-06-2003	31-12-2002	09-12-2002	279	01-01-2026

c. **Fungsi Konversi**

Jawaban :

Welcome Page | TP Modul 03

Worksheet | Query Builder

```
SELECT
nama,
TO_CHAR(tanggal_lahir,'DD-MM-YYYY') AS tanggal_string,
TO_CHAR(tinggi_badan) AS tinggi_char,
TO_NUMBER('12345') AS contoh_number,
TO_DATE('2024-05-20','YYYY-MM-DD') AS contoh_date
FROM mahasiswa;
```

Query Result

SQL | All Rows Fetched: 15 in 0.017 seconds

	NAMA	TANGGAL_STRING	TINGGI_CHAR	CONTOH_NUMBER	CONTOH_DATE
1	Aldi Prayoga	12-01-2003	170	12345	20-05-2024
2	Budiman Santoso	20-05-2002	168	12345	20-05-2024
3	Ciara Putri	15-07-2003	160	12345	20-05-2024
4	Dodi Nugroho	18-03-2001	172	12345	20-05-2024
5	Ekantha Putri	25-09-2002	158	12345	20-05-2024
6	Firman Abdul	11-11-2003	175	12345	20-05-2024
7	Gira Firmansyah	02-02-2002	162	12345	20-05-2024
8	Haryanto Kurniawan	30-12-2001	169	12345	20-05-2024
9	Indah Permata	09-04-2003	159	12345	20-05-2024
10	Jono Susilo	14-08-2002	173	12345	20-05-2024
11	Kiana Nanda	06-06-2003	161	12345	20-05-2024
12	Lukman Hakim	01-10-2001	174	12345	20-05-2024
13	Meylani Sari	21-03-2002	157	12345	20-05-2024
14	Nanda Hakim	17-09-2003	171	12345	20-05-2024
15	Oki Purwanto	05-12-2002	176	12345	20-05-2024

d. Fungsi Numerik

Jawaban :

Welcome Page | TP Modul 03

Worksheet | Query Builder

```
SELECT
nama,
ABS(-10) AS nilai_abs,
MOD(berat_badan,5) AS sisa_bagi,
FLOOR(tinggi_badan/3) AS floor_hasil,
POWER(2,3) AS pangkat,
ROUND(berat_badan/3,2) AS pembulatan,
CEIL(tinggi_badan/4) AS pembulatan_atas
FROM mahasiswa;
```

Query Result

SQL | All Rows Fetched: 15 in 0.008 seconds

	NAMA	NILAI_ABS	SISA_BAGI	FLOOR_HASIL	PANGKAT	PEMBULATAN	PEMBULATAN_ATAS
1	Aldi Prayoga	10	0	56	8	21,67	43
2	Budiman Santoso	10	0	56	8	20	42
3	Ciara Putri	10	0	53	8	16,67	40
4	Dodi Nugroho	10	3	57	8	22,67	43
5	Ekantha Putri	10	3	52	8	16	40
6	Firman Abdul	10	0	58	8	23,33	44
7	Gira Firmansyah	10	2	54	8	17,33	41
8	Haryanto Kurniawan	10	1	56	8	22	43
9	Indah Permata	10	4	53	8	16,33	40
10	Jono Susilo	10	2	57	8	24	44
11	Kiana Nanda	10	1	53	8	17	41
12	Lukman Hakim	10	4	58	8	23	44
13	Meylani Sari	10	2	52	8	15,67	40
14	Nanda Hakim	10	2	57	8	22,33	43
15	Oki Purwanto	10	3	58	8	24,33	44